Today

- Linear Systems of Equations
- Gaussian Elim.
- Partial Pivoting

$$2x_1 + x_2 = 4$$
$$-x_1 + x_2 = 1$$

Unkowns: $x_1, \ldots, x_n$

Linear Eqn.: $\sum_{j=1}^{n} a_j x_j = b$

no nonlinear functions of x_j
x^2, e^x, $\log(x)$, $\sin(x)$ etc.

System of m linear equations $\sum_{j=1}^{n} a_{ij} x_j = b_i \quad i = 1 \ldots m$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{m1} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Book: $[A]\{x\} = \{b\}$

Handwriting: $A\vec{x} = \vec{b}$

for now $m = n$

HW0 $\Rightarrow A \cdot x$

Lab

$$\underset{\text{known}}{A}\ \underset{\text{unknowns}}{x} = \underset{\text{known}}{b}$$

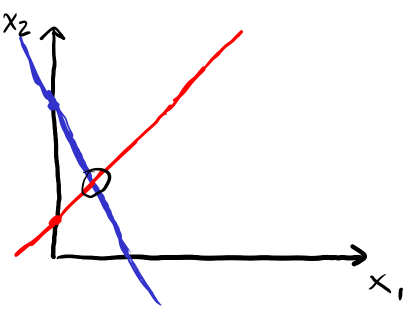
$2x_1 + x_2 = 4$

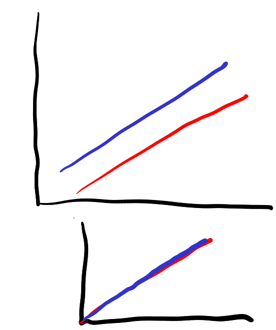
$x_2 = -2x_1 + 4$

$-x_1 + x_2 = 1$

$x_2 = x_1 + 1$

$\boxed{x_1 = 1, \; x_2 = 2}$





Example:

$$\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$3x_1 - x_2 = 0 \rightarrow 3x_1 - 3 = 0$

$\boxed{x_1 = 1}$

$2x_2 = 6$

$\boxed{x_2 = 3}$

$$2x_1 + 3x_2 + 4x_3 = 19 \quad (1)$$
$$4x_1 + 11x_2 + 14x_3 = 55 \quad (2)$$
$$2x_1 + 8x_2 + 17x_3 = 50 \quad (3)$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 4 & 19 \\ 4 & 11 & 14 & 55 \\ 2 & 8 & 17 & 50 \end{array}\right] \quad \begin{array}{c} f \\ -2\,(1) \\ -1\cdot(1) \end{array} \quad \left[\begin{array}{ccc|c} 2 & 3 & 4 & 19 \\ 0 & 5 & 6 & 17 \\ 0 & \boxed{5} & 13 & 31 \end{array}\right]$$

$$-1\,(2) \quad \left[\begin{array}{c} 2x_1 + 3x_2 + 4x_3 \doteq 19 \\ 0 \quad 5x_2 + 6x_3 \doteq 17 \\ 0 \quad 0 \quad \underbrace{7x_3 = 14}_{\boxed{x_3 = 2}} \end{array}\right]$$

$$2x_1 + 3\cdot 1 + 4\cdot 2 = 19$$
$$2x_1 = 8$$
$$\boxed{x_1 = 4}$$

$$5x_2 + 12 = 17$$
$$\boxed{x_2 = 1}$$

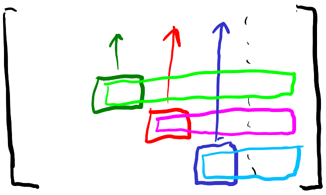
Eliminate □ with □
Eliminate □ with □
etc

Example:

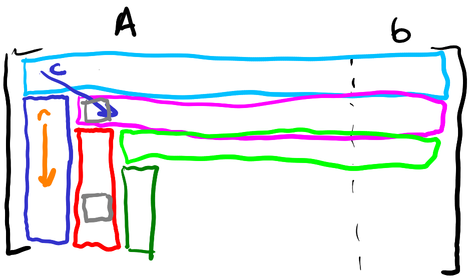
$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 9 & -2 \\ 2 & 1 & -1 \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ 9 \\ 2 \end{bmatrix}$$

$$x = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Back substitution



Solve for □ with □
Solve for □ with □
etc.



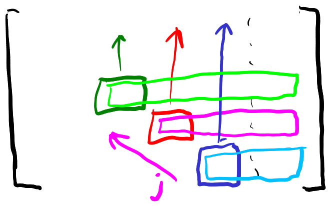
Elimination

```
function Gauss (A,b)
    n ← length(b)
    Aug ← [A | b]
    for c in 1:n
        for r in c+1:n
            f ← Aug[r,c]/Aug[c,c]
            Aug[r,c:end] ← Aug[r,c:end] - f·Aug[c,c:end]
```

Backsub

```
    x ← zeros(n,1)
    x[n] ← Aug[n:end] / Aug[n,n]
    for j in n-1:-1:1
        b_j ← Aug[j,end]
        x[j] ← (b_j − Aug[j,j+1:n]·x[j+1:n]) /Aug[j,j]
                     dot product
                  of all x's we've solved for
    return x
```

Back substitution



$$A = \begin{bmatrix} \boxed{0} & 2 & 3 \\ \boxed{4} & 6 & 7 \\ 2 & -3 & 6 \end{bmatrix} \qquad b = \begin{bmatrix} 8 \\ -3 \\ 5 \end{bmatrix}$$

$f \leftarrow \underline{Aug[1,2]} / \underline{Aug[1,1]}$ Divide by Zero

Solution: Partial Pivoting

```
function Gauss_Pivot(A, b)
    n ← length(b)
    Aug ← [A | b]
    for c in 1:n
        best ← argmax(abs(Aug[c:end, c])) + c-1
        Aug([[best, c], :] ← Aug[[c, best], :]
        for r in c+1:n
            f ← Aug[r,c]/Aug[c,c]
            Aug[r, c:end] ← Aug[r, c:end] - f · Aug[c, c:end]

    x ← zeros(n,1)
    x[n] ← Aug[n:end] / Aug[n,n]
    for j in n-1:-1:1
        bj ← Aug[j,end]
        x[j] ← (bj - Aug[j,j+1:n] · x[j+1:n]) / Aug[j,j]
                    dot product
                    of all x's we've solved for

    return x
```